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The volume of solid obtained when the region under the curve x=y over the interval [1,4] is revolved about the y-axis is

V=∫14πy2dy

If the upper and lower limits for the definite integral are the same, then ∫aaf(x)dx=If the upper and lower limits for the definite integral are the same, then ∫aaf(x)dx=

Zero

The value of ∫12dx=

1

Let R be the plane region bounded above by a continuous curve y=f(x) below by the x-axis and on the left and right, respectively, by the lines x=a and x=b the volume of the solid generated by revolving R about the y-axis is given by _____.

V=∫ab2πxf(x)dx

The volume by the Washers generated by revolving the region around y-axis is given by the formula ----------

∫abπ([u(y)]2+[v(y)]2)dy

The method of slicing by integration is used for finding ---------

Volume

EVERY continuous function on an interval has an anti-derivative ……

In order to fully determine the anti-derivative of a function f (F(x)), we must have.....

If the integral off(x) = x from x = 1 to x = 3 is 4, then the integral of f(x) = 10x from

x = 1 to x = 3 is _____.

4

IfF(x)istheanti−derivativeoff(x)on[a,b],i.eF/(t)=f(x),thenIfF(x)istheanti−derivativeoff(x)on[a,b],i.eF/(t)=f(x),then

∫baf(x)dx=F(b)−F(a)

Ifthecurveover[a,b]isrevolvedabouty−axis,thenthevolumeiscalculatedbytheformula−−−−−−−

∫abπ[f(y)]2dy

Integral of (1-2x) from [0,1] is ……..

0

The volume of the solid bounded by planes y=a and y=b with cross-sectional area A(y) perpendicular to the y-axis is

V=∫abA(y)dyV=∫abA(y)dy

First fundamental theorem of calculus gives the definite integral of a ..... function on a given closed interval in a quick way.

Continuous

x44−14=___

∫1xt3dt

We can break up definite integrals across a sum or difference

∫baf(x)±g(x)dx=∫abf(x)±g(x)dx=

As

∫baf(x)dx±∫bag(x)dx∫abf(x)dx±∫abg(x)dx

If the solid is revolved around the y-axis and generates a solid with a circular cross section of radius g(y) at y. Then the area of this cross section is

π[g(y)]2π[g(y)]2

∫abf(x)dx=_____.

−∫baf(x)dx−∫baf(x)dx

EVERY continuous function on an interval has an anti-derivative ……

On that interval

In integration of f(x)=x(x^2 +1)^3 from x=0 to x=2 by substitution method, we take u=x^2+1 then du= .....

2xdx

If the upper and lower limits for the definite integral are the same, then ∫aaf(x)dx=If the upper and lower limits for the definite integral are the same, then ∫aaf(x)dx=

negative integer

The volume of the solid generated by the region enclosed between y=x−−√y=x x=1 , x=3 and x-axis is resvolved by y-axis.Which of the following equation gives the volume of solid by cylindrical shell ____.

V=∫132πxx−−√dxV=∫132πxxdx

The integral of f(x) =sin(x) from x=0 to x=pi is

0

Theregionboundedbythecurvesy=x−−√,y=1andx=4Thereregionboundedbythecurvesy=x,y=1andx=4

½

If f(x) =3x^2 then F(x) (antiderivative of f) will be

6x

If the upper limit of Definite Integral is equal to its lower limit, then the value of Definite Integral will be _____.

Zero

The volume of a cylinder is the area of a cross section of the cylinder multiplied by the _____ of the cylinder.

Height

Findtheareabetweeny=xandy=−x(x−4)Findtheareabetweeny=xandy=−x(x−4)

None of these